



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC601 - OPERATIONS RESEARCH AND PROJECT MANAGEMENT

20EC505 - OPERATIONS RESEARCH

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define operations research. Explain the different phases of Operations Research.	10	L*2	1	1
	b) A company owns two mills A and B which have different production capacities for low, high and medium grade oil. The company enters into a contract to supply oil to a firm every week with 12, 8, 24 barrels of each grade respectively. It costs the company Rs.1000 and Rs.800 per day to run the mills A and B. On a day A produces 6, 2, 4 barrels of each grade and B produces 2, 2, 12 barrels of each grade. Formulate an LPP to determine number of days per week each mill will be operated in order to meet the contract economically.	10	L4	1	2
2.	a) Solve the following LPP using graphical method Maximize $Z = 80x_1 + 55x_2$ Subject to $4x_1 + 2x_2 \leq 40$ $2x_1 + 4x_2 \leq 32$ $x_1 \geq 0, x_2 \geq 0$	12	L3	2	1
	b) What are the applications of operations research?	8	L2	1	1
3.	a) A Manufacturer of furniture makes two products chairs and tables. Processing of this product is done by two machines A and B. A chair requires 2 hours on machine A and 6 hours on machine A and 6 hours on machine B. A table requires 5 hours on machine A and no time on machine B. There are 16 hours of time per day available on machine A and 30 hours on machine B. Profit gained by the manufacturer from a chair and table is Rs.2 and Rs,10 respectively. What should be the daily production of each of two products? Solve using graphical method.	12	L4	2	2
	b) Explain the following with respect to graphical method. i) Feasible solution ii) Feasible region iii) Optimal solution iv) Unbounded solution	8	L2	2	1
Unit – II					
4.	a) Solve the following LPP using simplex method. Maximize $Z = 5x_1 + 3x_2$ Subject to $3x_1 + 5x_2 \leq 15$ $5x_1 + 2x_2 \leq 10$ $x_1 \geq 0, x_2 \geq 0$	10	L3	3	1
	b) Solve the following LPP using Big-M method Minimize $Z = 2x_1 + 3x_2$ Subject to $x_1 + x_2 \geq 5$ $x_1 + 2x_2 \geq 6$ $x_1 \geq 0, x_2 \geq 0$	10	L3	3	1

P.T.O.

5. a) Solve the following LPP using Two-phase method.

$$\text{Maximize } Z = 3x_1 - x_2$$

$$\text{Subject to } 2x_1 + x_2 \geq 2$$

$$x_1 + 3x_2 \leq 2$$

$$x_1 \leq 4,$$

$$x_1 \geq 0, x_2 \geq 0$$

- b) Write a note on degeneracy in transportation problem.

12 L3 3 1
8 L2 4 1

6. a) Determine the initial basic feasible solution to the following transportation problem using i) North-west corner rule ii) Vogel's approximation method.

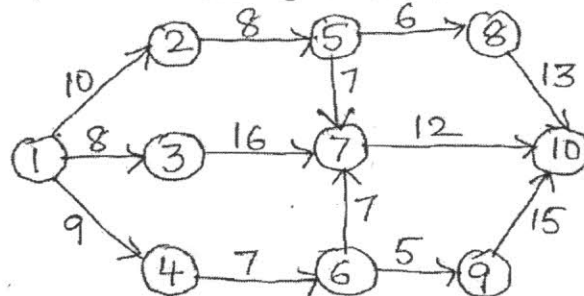
Warehouse →	W ₁	W ₂	W ₃	W ₄	Factory capacity
Factory ↓					
F1	19	30	50	10	7
F2	70	30	40	60	9
F3	40	8	70	20	18
Warehouse requirement	5	8	7	14	

- b) Explain the steps involved in Two-Phase method.

12 L3 4 1
8 L2 3 1

Unit – III

7. a) Explain the steps involved in project scheduling by CPM.
b) Determine the early start and late start in respect of all node points and identify critical path for the following network.



8. a) Explain the steps involved in the Hungarian Method to solve the Assignment Problem.
b) Solve the following instance of the Assignment problem.

	a	b	c	d	e
A	160	130	175	190	200
B	135	120	130	160	175
C	140	110	155	170	185
D	50	50	80	80	110
E	55	35	70	80	105

12 L3 5 1

8 L2 5 1

12 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
